

Where Occupations Move: Interstate Migration by Occupation in the United States, 2005 to 2024

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Extended summary

Interstate migration in the United States has been measured extensively for persons and rarely for occupations. That is an awkward gap, because most of the rules which make moving difficult attach to jobs, and because a state deciding whether to train workers or to recruit them needs to know what share of its workforce in a given occupation arrives from outside, and who those arrivals are.

This paper builds bilateral interstate migration flows by occupation from nineteen vintages of American Community Survey microdata covering 2005 to 2024, 59.8 million person records across 51 jurisdictions and 475 harmonised civilian occupation codes, and uses them to measure the share of each state's occupational workforce that arrived from another state within the year. Among the 400 occupations with enough records to measure it, that share ranges close to fifteenfold, from 0.66 per cent for tool and die makers to 9.70 per cent for sailors, and the design-based confidence intervals at the two ends of the distribution are an order of magnitude apart.

Three results follow from the panel. The first is that interstate migration delivers young workers and delivers them symmetrically: arrivals and leavers both average 34 years of age in the typical state and occupation cell, the workforce they join or leave averages 41, and in every one of the 82 occupations with enough arrivals to measure the incomers are younger than the people already doing the job. The second comes from a gravity model estimated by Poisson pseudo-maximum likelihood, which puts the elasticity of flows with respect to the origin-destination wage differential at 0.114 against a distance elasticity of -0.72 , so that a ten per cent pay advantage buys about one per cent more migration while distance carries some six times the weight. The third is that the deviations from that benchmark reproduce occupational concentrations which were never supplied to the model, placing Washington state at 2.82 times its predicted intake of software developers, the District of Columbia at 3.64 times its predicted intake of lawyers, and Michigan at 4.16 times its predicted intake of mechanical engineers.

The measures are tested against the Occupational Employment and Wage Statistics, an establishment survey collected independently of the household microdata from which every migration figure here is built. Ten hypotheses were specified before the results were examined and all ten are reported, with p values corrected across the battery; three survive that correction, in that the gravity residual tracks the location quotient, external dependence predicts employment growth and net inflow predicts employment growth, while the three wage tests return coefficients of -0.0004 , -0.0014 and $+0.0001$ against standard errors between 0.004 and 0.007, which is precise enough to exclude a wage relationship of any size a policy could act on. Two unrelated collections there-

fore place interstate migration where employment is expanding, and neither finds a measurable connection to where pay is rising.

1 Introduction

Roughly 2.3 per cent of employed Americans crossed a state line in a given year over the period studied here, and the stability of that figure across two decades conceals a distribution that is wide throughout. Tool and die makers replenish two thirds of one per cent of their number from other states each year while sailors replenish close to a tenth; farmers, postal workers, bus drivers and highway maintenance workers sit within half a percentage point of the floor, and pilots, air traffic controllers, aircraft mechanics and physical scientists replenish between six and nine per cent of theirs. A policy instrument aimed at labour mobility acts somewhere inside that distribution, which has not previously been measured at occupation detail across a panel of this length.

The absence follows from what the available data could support. Migration has been studied either at a level of aggregation which suppresses occupation entirely, from Molloy et al. (2011) on the national decline to Kennan and Walker (2011) on the individual decision, or one occupation at a time, as in the studies of individual licensure compacts. Neither approach yields the intermediate object constructed here, bilateral flows by occupation, state and year, over a period long enough to observe change in them.

The question a workforce authority puts to such data is narrower than the academic one. It asks whether the people filling a given occupation locally were trained locally or arrived from elsewhere, whether the arrivals are the ones needed, and whether recruitment or training is the instrument more likely to work, and all three require bilateral flows resolved by occupation and by age.

The claims made here are descriptive, and no causal effect of any policy is identified. The contribution is a set of measures constructed from public data, reported with the uncertainty the survey design implies, and tested against a source from which they were not built.

Section 2 describes the data and the construction of the flow panel, Section 3 reports the scale of external dependence across occupations, and Section 4 turns to who moves; Sections 5 and 6 estimate the gravity model and use departures from it to locate occupational agglomeration, and Sections 7 and 8 test the measures against outside data and set out the limitations.

2 Data and construction

Source and coverage

The flows are built from IPUMS USA extracts of the American Community Survey one-year files for 2005 to 2024 (Ruggles et al., 2024), 59.8 million person records in total. There is no 2020 vintage, because the Census Bureau suspended one-year collection in that year, so the panel runs to nineteen vintages with a gap in the middle, and any calculation crossing 2019 to 2021 measures a two-year change. Occupation is coded on the OCC2010 basis, which is the IPUMS harmonisation, and the four military codes are set aside because service members relocate on posting. That leaves 475 civilian occupations, of which 432 appear in every vintage; the remainder enter or leave at the

2010 and 2018 revision boundaries, where the harmonisation absorbs most but not all of the change in the underlying classification, and every occupation named in this paper is among the 432.

A worker is counted as an interstate mover if the state of residence one year previously is a valid state other than the current one. The rule selects a strict subset of the survey's own interstate migration flag, since no record is a mover by the rule alone. What it does discard is the 0.6 per cent of movers who carry the flag without a recorded origin state, a flow being unassignable to a pair of states without one.

Flows and zeros

Flows are counted for every origin, destination, occupation and year, and the zeros are retained. The estimation grid is restricted to occupation-years whose wage is measured from at least thirty non-moving workers at both ends of the corridor, which leaves 4.02 million cells across 400 occupations and 2,550 state pairs, and of those cells 91.6 per cent record no movers, and a corridor carrying nobody in a given year is informative about the difficulty of that move. Dropping the zeros, which taking logarithms of flows requires, would estimate the model on a selected sample of large corridors and would introduce a bias whose direction cannot be signed in advance.

Wages

The wage panel gives the mean log hourly wage by occupation, state and year, built from wage and salary workers who did not cross a state line in the year, and that restriction is necessary because the destination wage would otherwise absorb the selection the model exists to detect: workers who move in response to a good offer earn more on arrival, so an elasticity estimated on wages including theirs would be partly a construction of the measurement. Movers are 2.3 per cent of the workforce, so little is lost by excluding them. Hourly pay is computed from annual wage income, usual weekly hours and weeks worked, which removes the arbitrary threshold on hours that an annual measure requires and corresponds to the rate at which a job is actually offered. A cell mean built from thirty or more workers is still partly noise, and the reliability of the wage variable, defined as the share of its observed variation that is not sampling variance, is computed and carried into the estimates below.

Distance and standard errors

Distance is the great-circle distance between state centres of population, which places the origin of Nevada near Las Vegas where its population is concentrated and away from its sparsely inhabited geographic centre, and contiguity is derived from the Census county adjacency file.

Standard errors on the sourcing measures are computed by Taylor linearisation on the survey's strata and clusters, which the ACS design supports without collapsing: there are 2,462 strata in the final year, never fewer than 166 clusters in any stratum of any year, and no singletons. The median relative standard error across occupations is 6.1 per cent, and 345 of 475 occupations fall below 10 per cent.

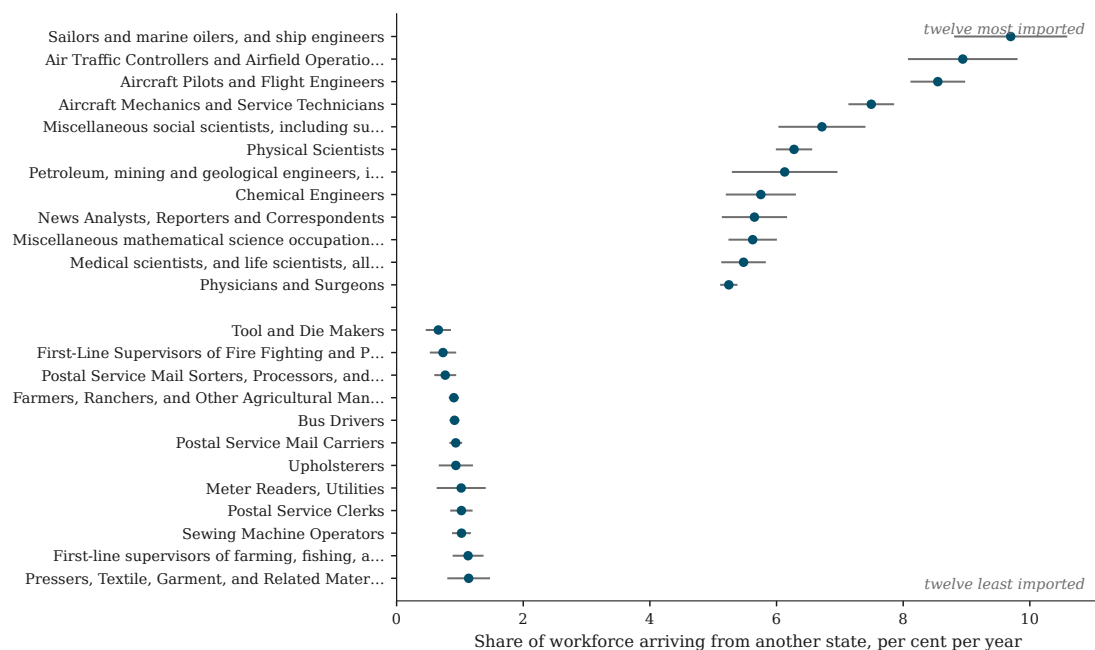


Figure 1: External dependence by occupation, pooled 2005 to 2024. Points are the share of the employed workforce arriving from another state within the year; bars are 95 per cent intervals from Taylor linearisation on the survey design. Occupations with fewer than 5,000 person records are omitted.

3 The scale of external dependence

External dependence is the share of a state's employed workforce in an occupation that arrived from another state within the year. Pooled across states and vintages, and among the 400 occupations with at least 5,000 person records, it ranges from 0.66 per cent for tool and die makers to 9.70 per cent for sailors and marine oilers, a spread of close to fifteenfold whose ends remain an order of magnitude apart at the bounds of their confidence intervals, with the most imported occupation bounded below at 8.80 per cent and the least bounded above at 0.86. Figure 1 reports the two ends of the distribution.

Occupations at the top of the distribution either serve networks which span states by construction, as sailors, pilots, air traffic controllers and aircraft mechanics do, or draw on a national market for scarce training, as physical scientists, chemical engineers and physicians do. Occupations at the bottom are attached to a fixed local asset such as a farm, a postal route, a stretch of highway or a municipal fire service. Licensure organises the distribution poorly, which is worth stating because licensure is the barrier the mobility literature most often names: physicians and veterinarians are among the most heavily licensed occupations in the country and sit near the top of the ranking, while postal workers require no licence at all and sit near the bottom. Through the middle of the distribution the ordering bears no visible relation to the intensity of licensing, and a reader who covered the labels and tried to guess which occupations were licensed from their position alone would do no better than chance.

Summed over the country, arrivals and departures are the same people counted twice, so the national gross rate of 4.1 to 5.4 per cent a year is simply double the arrival rate and carries no separate information. The distinction begins to matter at state level, where a net figure and a gross



Figure 2: Left: annual external dependence by age band, all occupations pooled. Right: mean age of arrivals against mean age of the existing workforce, one point per state and occupation with at least 100 arrivals and 100 departures. Points below the diagonal indicate arrivals younger than the workforce they join.

figure can diverge without limit: the District of Columbia replaces close to a fifth of its workforce across state lines each year while its net balance is approximately zero, which a measure built on net flows alone would record as no activity at all, in a city where a fifth of the people at work in any year were living somewhere else the year before.

4 Migration and the life cycle

That migration is concentrated among the young is long established (Rogers and Castro, 1981). The occupational resolution adds two things to it: the pattern holds in every occupation measured, and the flow is symmetric in age between arrivals and departures.

Annual external dependence falls monotonically with age, from 4.54 per cent among workers under 25 to 0.98 per cent among those aged 55 and over, a gradient of four and a half to one shown in the left panel of Figure 2. That arrivals and leavers have the same mean age nationally, 34.1 years against 34.1, is an identity, since every interstate arrival is somebody else's departure; what is not an identity is that the equality holds cell by cell. Across the 1,130 state and occupation cells with at least a hundred arrivals and a hundred departures, the median gap between the mean age of arrivals and the mean age of leavers is -0.2 years and two thirds of cells fall within two years of each other, against a workforce whose mean age is 41.5, so far as the cells can show the channel exchanges workers of one age for workers of the same age.

All but two of the 1,130 points in the right panel of Figure 2 lie below the diagonal, and the two exceptions are both waiting staff, an occupation whose workforce is itself young. Across the 82 occupations with at least 2,000 observed arrivals the incomers are younger than the workforce they join without exception, from 2.9 years younger for waiters to 11.9 years for aircraft mechanics,

Table 1: Wage, distance and contiguity elasticities of interstate migration flows, Poisson pseudo-maximum likelihood on 4.02 million cells, standard errors clustered by state pair. Distance and contiguity are absorbed once state pair effects enter.

Fixed effects	Wage	<i>p</i>	Log distance	Contiguity
Origin, destination, year	0.075 (0.022)	0.0006	−0.727 (0.028)	0.671 (0.048)
Origin-year, destination-year, occupation	0.109 (0.026)	<0.0001	−0.722 (0.028)	0.681 (0.048)
+ state pair	0.108 (0.026)	<0.0001		
+ occupation-year	0.114 (0.026)	<0.0001		

11.0 for physicians and 10.8 for postsecondary teachers. Interstate migration therefore leaves the age composition of a state’s occupational workforce roughly where it found it, exchanging early-career members for early-career members drawn from elsewhere.

A state facing a shortage of experienced workers in an occupation cannot recruit its way out of it through interstate migration, because the channel does not carry experienced workers in any occupation at any level of pay, which leaves training and early-career retention as the available instruments.

5 The gravity model

Specification

Flows are estimated by Poisson pseudo-maximum likelihood, which accommodates the zeros and the heteroskedasticity that a log-linear specification mishandles (Santos Silva and Tenreyro, 2006), on 4.02 million origin by destination by occupation by year cells. Estimating one occupation at a time is not viable, and the attempt returns a null which is an artefact of the design. Within a single occupation the wage differential is identified only from movement in a state by occupation cell mean over time, and a reliability calculation attributes 54 per cent of that movement to sampling noise, so the coefficient is attenuated to roughly half its size against standard errors too wide to detect it. Pooling occupations identifies the elasticity from variation across occupations within a state pair and year, whose reliability is 0.76 before the fixed effects are applied, and it permits origin-by-year and destination-by-year effects which absorb the time-varying multilateral resistance that would otherwise sit in the error term (Anderson and van Wincoop, 2003; Bertoli and Fernández-Huertas Moraga, 2013; Head and Mayer, 2014).

Wage and distance elasticities

Table 1 reports the wage elasticity across four specifications of increasing severity, with standard errors in parentheses. The estimate is stable and it rises as effects are added, which indicates that omitted multilateral resistance biased the looser specifications towards zero. The preferred estimate survives origin-by-year, destination-by-year, occupation-by-year and dyad effects together, so it is identified from the way wage differentials vary across occupations within one state pair in one year.

An elasticity of 0.114 means that a destination paying ten per cent more than the origin draws about one per cent more migration in that occupation, and that a state seeking ten per cent more

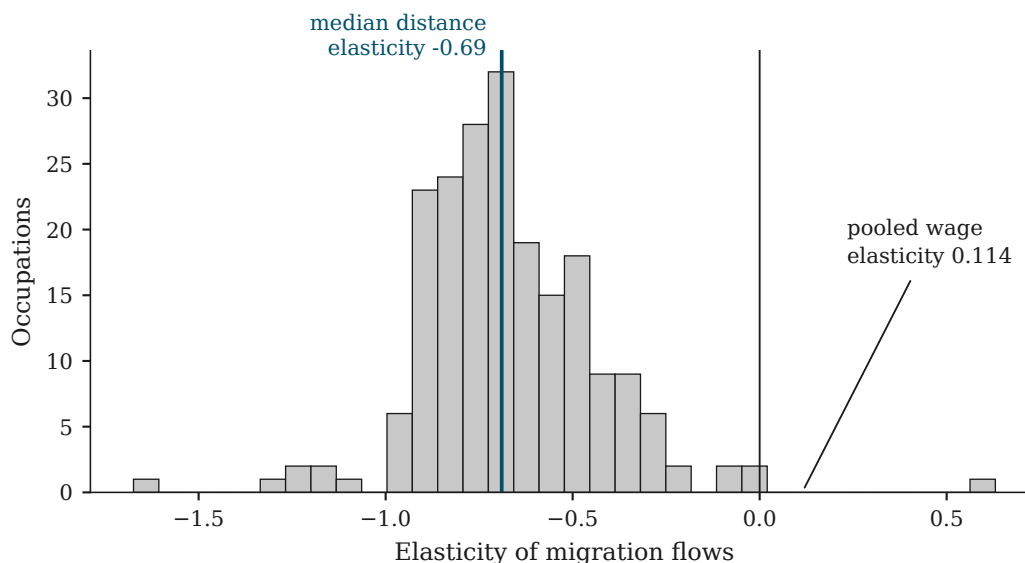


Figure 3: Distribution of the distance elasticity across 203 occupations estimated separately, with the pooled wage elasticity marked for scale.

arrivals through pay alone would have to more than double the occupation's wage. Dividing by the reliability of the wage variable raises the estimate to 0.15, and since the fixed effects remove part of the signal in the wage differential and none of the sampling noise, the reliability of the variation that survives them is lower than 0.76, so 0.15 is a floor under the corrected figure.

The distance elasticity is -0.72 in the pooled model, some six times the wage elasticity, and unlike the wage elasticity it can be estimated occupation by occupation, because it is identified from variation across 2,550 state pairs and not from movement over time. Estimated separately for the 203 occupations with at least a hundred populated corridors it has a median of -0.69 , an interquartile range of -0.81 to -0.53 , and a central ninety per cent running from -0.96 to -0.29 , so the friction varies about threefold across the body of the distribution; it is distinguishable from zero in 93 per cent of occupations, and the tails beyond that range, where editors sit near zero and flight attendants near -1.7 , carry standard errors wide enough that they are better read as noise than as extremes. Within the body a coefficient near -0.3 belongs to an occupation recruited nationally and one near unity to an occupation whose recruitment beyond the neighbouring states returns very little. Figure 3 places the pooled wage elasticity against that distribution.

Turnover and net balance

Occupations differ twentyfold in the share of their workforce crossing a state line each year in either direction, from 1.29 per cent for tool and die makers to 25.52 per cent for avionics technicians, and very little of that traffic accumulates on either side. Nationally it cannot, since the arrivals and the departures are the same people, so the question is put at state level: across 9,775 state and occupation cells, the absolute net balance comes to 14.5 per cent of the gross traffic, and that ratio falls as the traffic rises, from 24 per cent in the least mobile quartile of occupations to 15 per cent in the most mobile. Aircraft pilots move 17.1 per cent of their workforce across state lines each year while the mean absolute state balance in them is 1.2 per cent, under a tenth of the gross; aircraft



Figure 4: Annual crossing rate against the employment-weighted mean absolute net balance across states, one point per occupation with at least ten state cells, marker area proportional to employment. The grey diagonal is where every arrival would add to the stock; the dashed line is the pooled ratio of net to gross.

mechanics move 15.0 per cent against a mean absolute balance of 2.4. Figure 4 shows the pattern across all occupations, with the net balance held to a narrow band well below the gross at every level of movement, and since sampling noise enters the absolute net balance and never leaves it, the 14.5 per cent is an upper bound on how much of the traffic actually accumulates.

The distinction matters for the instrument a state can use, since in a high-turnover occupation each worker attracted is matched by one lost, so that additional recruitment replaces the outflow without raising the stock and retention becomes the binding constraint, while in a low-turnover occupation the channel is too narrow to recruit through at all and the workforce has to be produced locally.

6 Departures from the benchmark

Once distance, contiguity, wage differentials and everything constant about each state in each year have been accounted for, the model still leaves a systematic component in the state by occupation cells. Figure 5 plots observed against predicted arrivals.

The residual reproduces occupational concentrations which were not supplied to the model. Washington state draws 2.82 times its predicted intake of software developers and California 2.03, the District of Columbia 3.64 times its predicted intake of lawyers, Michigan 4.16 times its predicted intake of mechanical engineers, and New York 2.76 times its predicted intake of securities and financial services agents. Under-supply appears with equal clarity, with South Carolina, Tennessee and Kentucky drawing between 0.28 and 0.44 of their predicted intake of software developers. California is the instructive case, since it draws twice its predicted software developers and half its

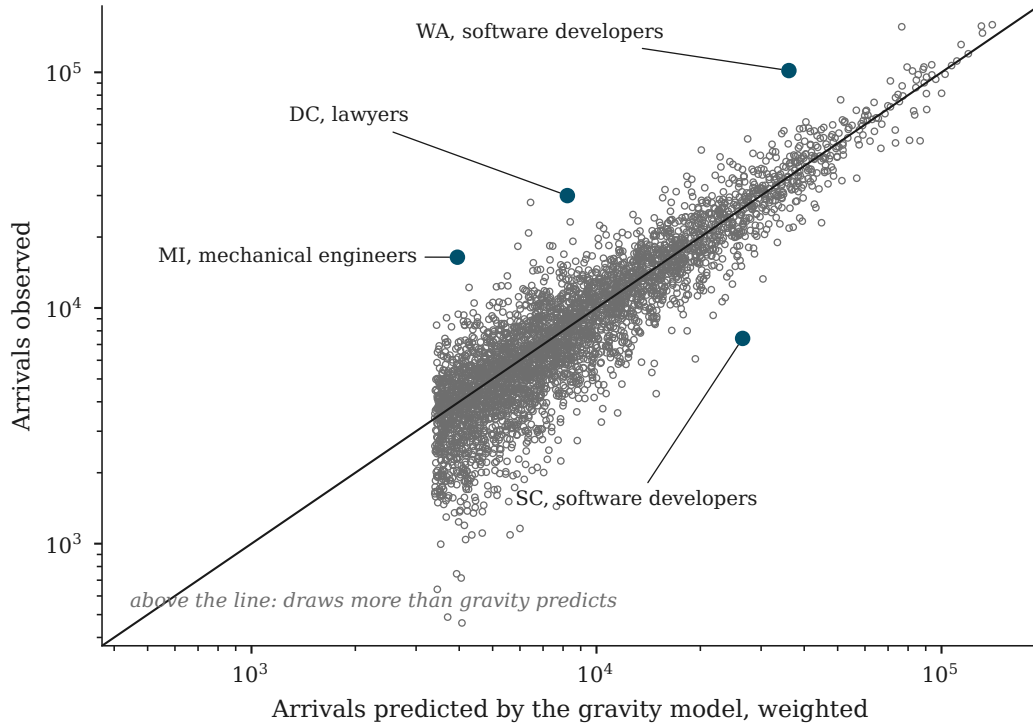


Figure 5: Observed against gravity-predicted arrivals by state and occupation, pooled 2005 to 2024, log scales. Cells below roughly thirty expected sampled movers are suppressed.

predicted home health aides, which is the pattern a high cost of living would produce by sorting arrivals on what they earn.

Deviations are judged against Poisson sampling variation, because a ranking on the raw ratio surfaces only small-cell artefacts: the standard deviation of the ratio falls from 0.39 in cells below 5,000 expected arrivals to 0.20 in cells above 50,000. On the noise-adjusted measure, 43 per cent of cells exceed a z of 2, against the 5 per cent that sampling variation alone would produce.

The concentrations are also stable over the period. The residual was aggregated into four multi-year windows on a balanced panel, and the correlation between the 2005 to 2009 and the 2021 to 2024 positions is 0.543. The noise-corrected spread of the residual is flat across those four windows, at 0.263, 0.276, 0.257 and 0.275. The geography of occupational concentration does not, on this evidence, measurably move across twenty years which contain a financial crisis, a pandemic and a rapid expansion of remote work.

7 Testing the measures against outside data

Every measure above is built from household survey records, and the Occupational Employment and Wage Statistics is collected from employers, so agreement between the two carries information that a comparison within a single source would not. Ten hypotheses were specified before the results were examined, all ten are reported, and p values carry a Benjamini-Hochberg correction across the battery (Benjamini and Hochberg, 1995). Regressions are weighted by employment and clustered by occupation and by state.

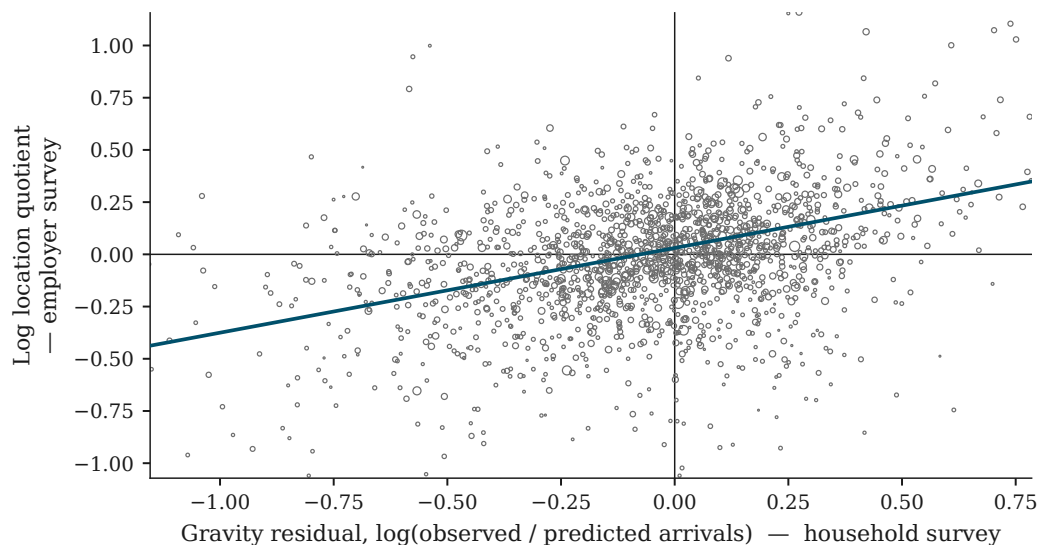


Figure 6: The gravity residual, from household survey microdata, against the location quotient, from the establishment survey. Marker area proportional to employment; the line is the employment-weighted fit.

Three of the ten survive correction, and the strongest of them is the gravity residual, which tracks the location quotient: one standard deviation of the residual is associated with a location quotient 13 per cent higher, with an adjusted p of 4.4×10^{-7} . That is the result shown in Figure 6, and it establishes that the residual carries a real signal about occupational concentration. External dependence predicts subsequent employment growth at 0.044, and net inflow predicts it at 0.029.

The wage tests return nothing, whether the predictor is the gravity residual, external dependence or net inflow: the relationship with five-year wage growth is -0.0004 , -0.0014 and $+0.0001$ respectively, against standard errors between 0.004 and 0.007, which is enough precision to treat all three as estimated zeros. The consequence is a direct one, in that a state drawing materially fewer workers than the benchmark predicts sees no faster wage growth, so the residual cannot be read as a shortage signal and is not presented as one here. A tenth result points the same way, with states paying above the national rate for an occupation importing very slightly fewer of them, wrong-signed and not significant after correction, which is the sign a cost of living effect would carry.

8 Limitations

Occupation and age are recorded at the time of survey, after any move has taken place. An arrival counted in an occupation either practised it before moving or entered it on arrival, and the data cannot separate the two cases. For occupations with long training requirements the ambiguity is minor, while for easily entered occupations the sourcing measure is an upper bound on imported trained workers, and the claims made here are restricted accordingly.

Coverage across state and occupation cells is uneven, so that the results are secure nationally, by occupation, and for large combinations of state and occupation, while only 4,026 of 24,102 state by occupation cells reach thirty arrivals and thirty departures even when all nineteen vintages are pooled. A complete lookup across every state and every occupation is therefore not available from

these data and is not offered here, and cells below that threshold are suppressed and not interpolated. The absent 2020 vintage makes any calculation spanning 2019 to 2021 a two-year change.

Nothing estimated here identifies a causal effect: the residual locates where a state diverges from its benchmark and is silent on the reason, and the association between migration and employment growth runs in a direction these data cannot determine.

9 Conclusion

Interstate migration supplies occupational workforces through a channel that varies close to fifteen-fold in width across occupations, that is narrow in every one of them, and that carries workers of a single age range. The measures constructed here put numbers on both facts, report them with the uncertainty the survey design implies, and agree with an independent employer survey wherever agreement is testable.

Three of the results bear on a state deciding how to staff an occupation. Pay is a weak instrument, since a ten per cent advantage buys about one per cent more arrivals and the wage tests against outside data return nothing; recruitment cannot supply experience, since none of the 82 occupations large enough to measure draws arrivals older than the workforce they join; and in the occupations where movement is heaviest the arrivals and the departures cancel, so that the constraint on the stock is retention.

The deviations from the gravity benchmark are large, stable across two decades and so far unexplained, and the apparatus assembled here locates them without accounting for them, which is the usual position of a measurement paper and the one this paper is content to occupy. Occupational licensure compacts, which take effect between specific pairs of states on specific dates, supply variation of the kind that could account for part of them, and that is the direction the work takes next.

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